Unit 3 - Impact of IT
Assignment 3

Introduction

Scenario / Vocational Context: *Now that iTech Consultancy Company has begun its marketing phase, the Company needs to inform its potential clients about any changes to their company that may results when upgrading ICT systems. This could be in the form of changes to their premises and staff, also any associated risks that introducing new technology might cause. You will also need to advise potential clients of the improvements that can be made in terms of the types and quality of information that their businesses may generate and use.*

Task 1

Scenario One – Bella Pizza

Bella Pizza is a traditional Pizza restaurant on the High Street. They are often very busy and as such need to employ a lot of staff to take orders to the kitchen. Often orders are late or incorrect as the Chef gets overloaded or misunderstands what the staff have written for an order. Also, there is little to no digital audit trail to track what supplies and they need, and also which pizza are most profitable or need to put on special offers.

They have heard about self-service Tablet ordering systems, with which customers choose their food at their table via a Tablet, which directly links to a screen in the kitchen.

(U4:P3) Explain how organisations respond to information technology developments

Task Description -*If Bella Pizza choose to implement Self Service Tablets to their business, explain how they would need to respond or change in the following areas:*

Sales Process

If Bella Pizza were to have the implementation of the Self Service tablets, the business will benefit in the sales process due to the much faster and much more efficient method of sales of products. So, the business would have to get rid of the employees and replace them with the self service tablets. The reason for this is due to the fact that a customer would have to go to an employee to purchase the goods, and there are many factors which can prove that this is slower than a customer using a self service tablet to purchase. One of the significant reasons for this is due to the human error of the employees.

Marketing Strategies

If Bella Pizza were to have the implementation of the Self Service tablets, this will make the business and the staff prioritise strategies a lot better. With the ordering and delivery of products automated thanks to self service tablets, this means that other employees can focus on different areas of the business.

Purchasing strategies for their supplies

If Bella Pizza decide to have the implementation of the Self Service tablets. The business will have to change the purchasing strategies so that it would be much more profitable and advantageous to the businesses since it has the self service tablets. The reason for this is due to the fact that information of sold products get stored in the database of the business. This means that the purchasing strategy will be much more organised, making it a much easier task to manage compared to when the company had overloaded information. With the information being more controlled, the Chef will be able to control and understand overloaded information a lot better.

Automation

If Bella Pizza were to have the implementation of the Self Service tablets, it has brought the automation to where the business survives from, the ordering and production of pizza. The self service tablets have helped remove the daring task of getting information from all the customers, and help the staff focus on other tasks which will improve the quality of the business and the place itself. This will also make it a much easier task to manage compared to when the company had overloaded information. With the information being more controlled, the Chef will be able to control and understand overloaded information a lot better.

Customer Support

If Bella Pizza were to have the implementation of the Self Service tablets, these new developments have allowed access to customer support by the most efficient ways a customer can use such as going online and typing up a question onto the organisations website through the ‘contact us’ section, being able to send in emails or by simply looking up for an answer on the FAQs page. On the other hand, technological advancements have resulted to using less satisfactory customer services such as using automated self service tablets, and this can be quite frustrating for customers who have to go through with this service in order to get what they want. Using this system can be quite unpopular.

Financial Management

When Bella Pizza had the implementation of the Self Service tablets, this will make it from extremely difficult to bearable, due to the fact that the business was a very busy business. So the financial manager could then control all of the problems that was occurring in the business all at once, which had the possibility of causing stress to the financial manager and forced to prioritise and sacrifice tasks.

Staff training and redundancies

If Bella Pizza were to have the implementation of the Self Service tablets, it means that the staff will no longer need to have to be trained on how to interact with the customers and focus more on qualitative data. The reason for this is because the tablets will provide the customers need into the database, which can then be used to supply the customer for the product that the customer wishes.

Scenario Two – JerseyTechGurus.com

urus.com is a local Technology retailer. They supply, install and support clients all over Jersey with all their computing and technology needs. Currently they provide all their services from their shop and their website. As the business has grown, it has become a very busy shop. With shop and web sales happening in one location. Also, the technical support they are providing is taking up a lot of manpower, in terms of enquires and associated visits to clients or telephone assistance.

The company are considering online customer relationship management (CRM) software to be able to deal with enquiries and provide support. It is hoped that they can pool together enquiries received from their website and social media into one place. As such they will become much more networked and connected to the customers and each other in different departments.

(U4:P4) Explain how an organisation can manage risk when using IT technology

Task description - *If JerseyTechGurus.com choose to use CRM software such as Zendesk, they will be putting much of the business and customer service through online contact only. Explain how they may need to manage or prevent the following new risks to their business:*

Increased risks to their finances and money. Money coming in and going out securely

When using IT technology, there is a risk that the finances and money is vulnerable to an external interceptor. What I mean by this is that, there are various types of hackers, offenders who commit fraud and illegitimate competitors that can commit an illegal act. Since money is the reason that the business is still alive, this is a high priority to keep it safe. So, to counter any of these risks, the company will have to invest on a very hard to infiltrate security system to protect their finance and money details.

Communication hacks or customer data theft

When using IT technology, the data is also very vulnerable to attacks due to the amount of reliability the company has with the IT technology. The data is very vulnerable to any offender who is planning on hacking the data or infiltrate confidential company conversations. To reduce this risk, the company should invest in a good security system to make the interception a lot longer to take, making it very difficult to infiltrate the data.

Protection of their intellectual and digital property they make or own

When using IT technology, there are the copyright laws in which protects the company's own intellectual and digital property, protecting the creative work and unique aspects of the company itself. However, copyright laws can be broken by a competitor or user. If that were the case, then the company that had it copyrighted could give a warning to the offender, or maybe even sue them if the offender made a profit from the copyrighted material.

DDoS attacks

When using IT technology, there is a risk that the company may experience a DDoS attack. These attack are simple yet effective, and very hard to counter due to its simplicity. What I mean by this is that a DDoS attack is basically when a server gets overflowed with useless packets at overwhelmingly amount of packets per second in a prolonged amount of time. These DDoS attacks can only be fully countered if the offender's location has been found and that he is fined to compensate for the company's lost profits or to spend time in imprisonment.

Complete physical loss of access to their offices to work from. Where will they work from to keep the business running?

When using IT technology, it is recommended to have an external facility or an office space in which they have data backed up and a computer system to access their data. They should also have staff with tablets and laptops to be able to help customers away from their office from either the different office that I talked of before or from home. If there is no external facility or office, it would be a better idea to get a laptop or other portable devices which has access to the company data, under strict security, so it wouldn't be lost.

Scenario Three – New Horizon Care Services

New Horizon provide care and support services to people in Jersey with disabilities and/or learning difficulties. They pride themselves on being able to provide care and support to people in their own home or independent living environment. Currently they have a team of support workers that visit clients. The visits are recorded in paper records and are filed with the care manager's weekly back at the office. The managers have received complaints about support workers being late or not spending long enough on visits. This is not the fault of the carers as they work very hard, but it seems they may need to be organised and managed in a more efficient manner. In addition to this upon inspection from the Government it has been noted that the information being stored is not in line with Data Protection Laws.

They would like a more efficient and up to date system to use, as such they are interested in a mobile smartphone app for support workers, which tracks their location and length of visit. Which is linked to their offices via WIFI and 3G. Which can then be stored in secure databases for Managers and inspectors to access and use information as and when they need to.

(U3:P1) Explain how organisations use information

Task description - *If New Horizon Care Services implemented this IT development and therefore digitalised and networked all of their business systems, data and management information systems (MIS). They would want answers to **all** the following questions in relation to the data and information being managed on their systems:*

Explain what qualitative information they may generate

If New Horizon Care Services had the implementation of the mobile application, they can measure the qualitative data, which is data that cannot be measured in the same way as

quantitative, so it is presented in a narrative form. It can't be measured, it can only be observed with descriptions. For example a description or lengthy answer that describes the data such as 'Some people use the wheelchair but others use a walking stick'.

Explain what quantitative information they may generate

If New Horizon Care Services had the implementation of the mobile application, they can measure the amount of quantitative data, which is data that can be measured numerically and then proven as fact. It is measured in numbers. For example, numerical values and exact measurements in their organisation, such as '25% use a walking stick, 40% walk, 30% use the wheelchair and 5% are immobile'.

Give examples of their Primary information

When New Horizon Care Services had the implementation of the mobile application, the data gathered from the mobile app can consist Primary data, which is data that the organisation gathers and manipulates itself. It is original as it is collected by the data manager. For example, the primary data is gained via surveys in order to gain the data straight from the source.

Give examples of their Secondary information

If New Horizon Care Services gets the implementation of the mobile application, the data gathered from the mobile app can consist secondary data. This data is gained via the customer's feedback. For example, the data that can be collected may be how much time each care worker has on their shift.

How will they be able to monitor and control their business activities and care workers?

If New Horizon Care Services had the implementation of the mobile application, the software will contain a presentation of information on dashboards containing the key performance indications of a business from customer feedback, which is used to provide assurance and visibility of activity and performance. This information can be used by business operations to provide visibility, measurement and assurance of key business activities.

How will the Managers be able to identify patterns or trends in the business?

If New Horizon Care Services had the implementation of the mobile application, the data that the mobile app will provide will also consist of data of certain customer and staff reviews changing in direction and all have an overall, somewhat similar system. For example, the data from the feedback from the careers and visitors will have very similar points towards a certain aspect of the business, such as the food stand.

How will their ability to make business decisions change?

If New Horizon Care Services had the implementation of the mobile application, it will be possible if the business decided to make changes into business decisions. It all basically depends on how the data from the feedback from the employees and customers or the data

proves that there is a small strategy used that might increase profiting. For example, the data may showcase that investing in a new operating system for the computers will be more profitable than buying newer computers in general.

What advantages will they have over similar businesses that do not use this type of technology?

If New Horizon Care Services had the implementation of the mobile application, they will have a much higher advantage over other competitors that do not have this technology. The reason being is because it is a very sophisticated and efficient piece of software that is simple to use and contains a vast amount of useful business data within its database. For example, if 2 businesses had to find out how many they need to invest on wheelchairs, the one with this technology can just check the database and order “x” amount of wheelchairs. While the other business without this technology would have to make a questionnaire in order to gather information, which will take significantly longer.

(U3:P2) Discuss the characteristics of good information

Task description - *It is expected that the quality of information New Horizon Care Services will be able to use will be greatly improved, but you need to be able to discuss with them how the characteristics of their information becomes classed as good. Discuss how the following aspects will become better through the proposed changes:*

How will their information be considered more valid and reliable?

Having valid information that is proved correct and useful for the reader's knowledge is known as valid information. Invalid information would not be useful for a user as incorrect information would supplied, preventing any benefit to be made to the user’s knowledge. Having reliable information means that this is sources of information that can be relied on for correct information. If it isn't reliable, this source would not be completely relied on for all information, further research may be needed to back-up its reliability.

How will their information be more timely and efficient?

Having timely information means that the information given is required to have time given to it; steadily planning what should be written, structuring out the information properly so it is understandable and thoroughly re-checking it to ensure the information is correct, clear and concise indicates that the information has had good time and consideration given to it. Having the efficient detail means that the information should be detailed and not brief. What I mean by this is that, the more it is explained in depth, the better the chances of understanding the information. Information that isn't clarified may not be useful to the reader.

How will their information be more accessible and useful?

Having accessible information means that all information processed should be stored away somewhere which can, when in need of the information again, be easily accessible so minimal time is taken when looking for it. When the information is useful, this means that any information

you gather must be useful and relevant to what you require it for; irrelevant information not related to the subject you are looking into will not make it fit-for-purpose.

How will their information now be more cost-effective?

If the information is Cost-effective, this means that the information that has been invested on paying organisations to help acquire information about something, such as surveys for a business, must be worth the sum paid for it, so benefit is made from the information attained.

How will their information be more accurate and understandable by their staff?

Having accurate information can prove to be very important, because information that isn't accurate enough may not be useful to the reader as they need reliable and correct information. It can also lead to wrong decision making due to the incorrect information provided. Having understandable source of information means that the found requires assurance and identification that it is correct and proven valuable for the user; doubting its reliability and whether it is useful to the user can affect the user's confidence on the sources.

Task 2

(U3:M1) Illustrate the information flow between different functional areas

Task description - *Now that you have shown prospective clients some of their business systems responses and potential effects on their information systems, you will need to show them how the flow of information may look within their business.*

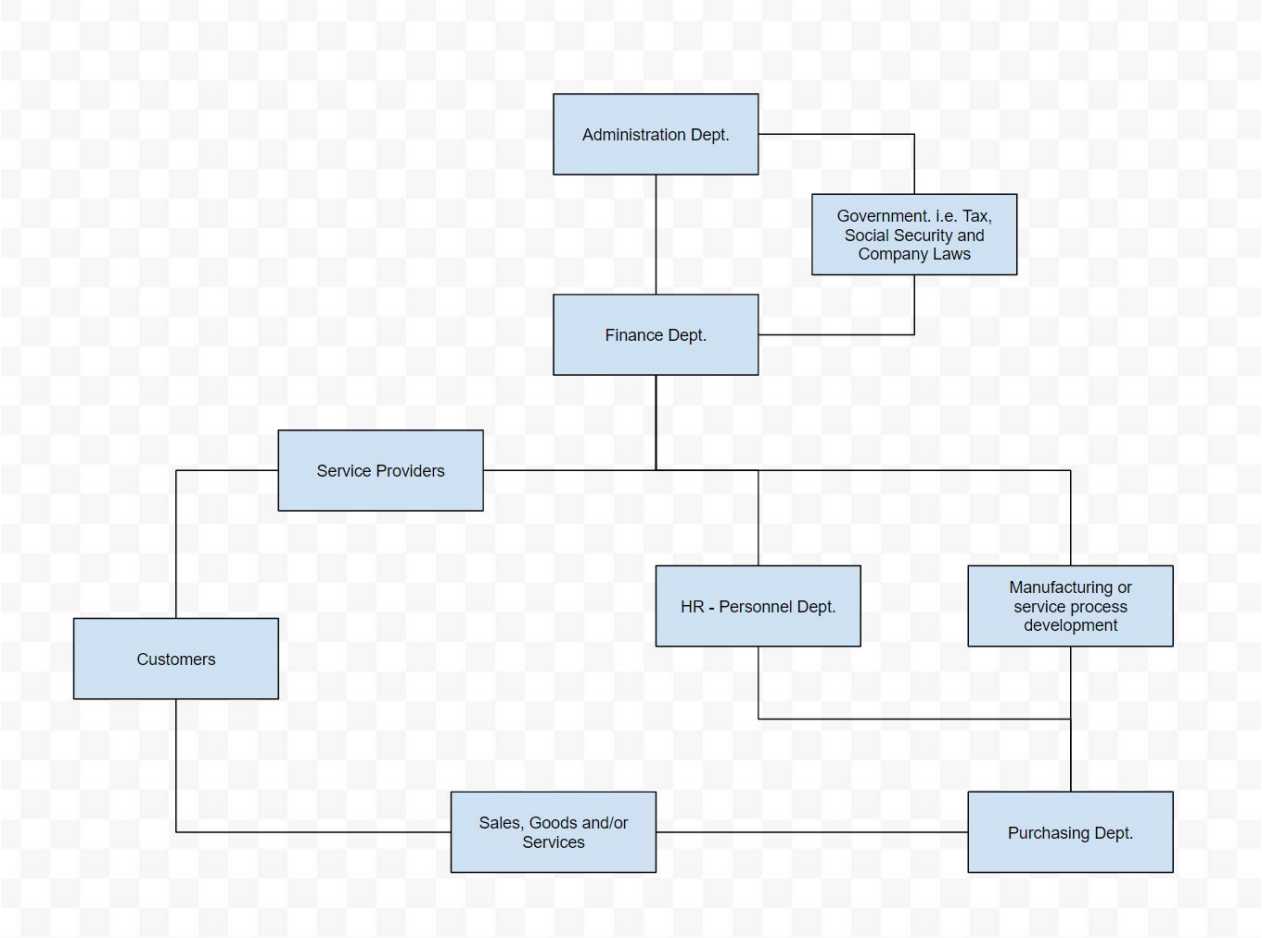
*You can choose any **one** of the above case studies to show the flow of information between functional areas of their businesses:*

Using a flow diagram show how the directional internal flow of information between the following functional areas of a business will be. You must show what information is being moved between the information system to each area:

- *Sales. Goods and or Services.*
- *Purchasing Dept.*
- *Manufacturing or service process development.*
- *Finance Dept.*
- *HR – Personnel Dept.*
- *Administration Dept.*

To your flow diagram add any external bodies that your information system may feed into our out of, considering:

- *Service providers*
- *Government. i.e. Tax, Social Security and Company Laws.*
- *Customers*



This flow diagram demonstrates the relationship flow in a business system, showing: the sales, goods and/or services; purchasing dept.; manufacturing or service process development; finance dept.; HR - personnel dept.; administration dept.; service providers; government i.e. tax, social security and company laws, and customers.

(U3:D1) Explain how an organisation could improve the quality of its business information

Task Description - *Using any of the case studies you have worked on above, explain in detail how they may be able to significantly improve the quality of the information generated and moved around the business. Your detailed explanation should demonstrate a good understanding of the notion of what good information is as per criteria (U3:P2) above:*

What additional IT developments could they consider to improve the quality of the business information

There are many IT developments in which could improve the quality of the business's information system. For example, the most basic form of technology, a computer, is a great way

to store information with a decent security and admin privileges, so that the data cannot be infiltrated and stolen from an external or internal offender.

However, It should be considered to invest in newer, additional IT technology such as getting a new Operating System, new computer systems or just new database software can really benefit the business due to the faster speeds in which the workload can be carried out.

The reason for this is because older systems are out of date, and newer ones tend to be a lot faster and more reliable than older ones. The reason for this is due to the fact that an updates system or software is basically an improved version of the old one where the bugs and other errors have been fixed.

When data is stored within a computer, it's usually in a database of some type. These databases provide validation rules which ensure all data entered is valid and any invalid data cannot be entered.

Validation rules should be specified so that each piece of data has a value, examples of validation rules will all be listed below to prove what additional IT development could do to improve the storage of data.

Data type, which consist of numbers, integers, text, date etc. This type of data can be used to measure the quantity of the products, employees, customers or other forms of data that can be measured using numeric or text values to create a good quality database for future improvement to the company.

There are also Minimums and Maximums, For example, "can there be a negative sales on a certain month?". This can be used to reinforce and to help make business decision which can be too risky for the CEO to carry out by himself.

It has to have values that are acceptable or unacceptable. For example is 'IRE' a code that is valid for Ireland? Data within a business can also contain codification rules which is basically a type of column of numbers, for example "what does it represent?" or "Are the invoice numbers randomly generated or does each segment of it represent something?"

Classification of the data is another validation rule that is used. An example of classification of the data would be the rules regarding what data belongs to what group. This will improve the quality of the data storage, because it help place data in different categories so that they can be accessed much more efficiently. It also reduces the error of getting the wrong information for the wrong business decision, which has the risk of being a catastrophe.

The Units of the data also improves the quality of the data because the units are basically the rules used so measurement units are consistent at all times, for instance: mm, cm, inch, and other methods of measurement. This can be used to measure how much facility space can be used to store useful products and systems.

However, the data will need to have an authority identification in order for it to be safe from anyone planning to tamper with the data. For example, In large and more complex systems the ultimate source of a data element is critical to data quality but may not be essential in simpler systems.

Frequency of updates are important so that data elements can be static and do not update often, however some dynamic data is event or transaction driven by business processes.

It is important to have a proper use and access description, depending on the population of users in a system, there may be data elements that may not be as equally accessible to users. User roles may be used.

Also, a data dictionary should provide information on what is included or excluded from data elements depending on the scope of a query although it doesn't have to include any underlying data and what may be used for calculations.

Also, another important aspect with the data storage is that the data that can be stored within a computer and many other electronic devices with storage space, can contain a data dictionary within them.

Basically, a data dictionary is a useful and valuable tool in the use of documenting data. It contains the rules regarding the quality of data. The following are some essential elements that should be included in a good data dictionary for the improvement of the quality of data.

All data elements should be identified in the data dictionary. There are many data elements that accumulate over time, mostly in informal data systems like spreadsheets but a lot of time and money can be saved on training and projects by making sure you have up to date lists of all data elements.

It is very important to have data element definitions, because if nobody can understand what an element is, it can lead to errors with interpretation which reduces the quality of data. Comprehensive definitions should be provided so the data elements can be understood.

Overall, improving the IT developments for the database is very important because it help the users understand how to make and use the data in IT in order for the business to improve its customer and product efficiency. Also, upgrading the older systems is important due to the fact that they run a lot faster and much more reliably, meaning that the workload and data can be carried out faster with a lot less errors and risks of losing work or data.

<u>How to make their business information more accurate and reliable?</u>

With the implementation of additional IT development, it is certain that the information used in the project and company could be improved to be more accurate and reliable. For example,

there are several ways in which this can be done. Firstly, the business can ensure that the data is valid and has the other characteristics of good information, they can ensure the type of data is suitable, and make sure that the data fits the purpose it is being used for.

For the information being valid, the validity of the information that is used in the project can greatly improve the quality of the information. When information is collected, it should be compared with other sources of information to ensure that it is correct and reliable before it is used. Information can be passed through the operational support system to be checked that it is valid for its purpose.

An example of where having valid information would improve the accuracy of information would be in the business team's bid for additional funding. When they were bidding for more money, the team did not have valid costs as some of the teams hadn't given them information on their expenditure. This meant the team had to make up data and give the mentor invalid information. If the data had been valid, the quality of the information given to the mentors would have been better.

For each different type of Data and Information, the type of data affects the quality of information; a business needs to be able to choose the right form of data for the purpose it is being used for. Firstly, the company has to decide whether qualitative or quantitative information is better. Then they need to decide whether to get the information from a primary or secondary resource.

Choosing the right type of information is important as it improves the quality of the information due to the data being more valid and reliable, especially if it is from a primary source. A combination of quantitative and primary data produces higher quality information as the data is numerical and collected by the company so is fit-for-purpose and valid.

If the marketing team used only one or the wrong type of information in their questionnaire, then the information that they used to market the classroom and improve the design would not be quality information. If they used the right types of information, then the quality of the information would improve and when used, it could help make the project better.

For the data to fit the purpose, information needs to be specific to its purpose in order to be considered quality information. Having a purpose for collecting the information allows the team to ensure that it is fit-for-purpose and from a valid source. Knowing the purpose of the information allows the company to decide.

With the data being valid or being up-to-date, if the company used information from a source that was either incorrect or was invalid, then the project could be negatively affected. For example, if one of the design teams used a website to find out the costs of their materials that was out of date, then it could impact the budget.

The prices on the website could be wrong and be higher than the team thought, this would mean that the project would go over budget. If the checked their sources to make sure that it was quality information they were using then this won't happen.

Overall, to make sure that the information of a business is accurate and reliable, the business has to make sure that the data is categorised properly and that other forms of information sources reinforce that of the original data, to make sure that it is accurate and reliable so it becomes advantageous to the company.

Ensuring the external sources of information are relevant

It is important the a business ensure that any external sources of information are relevant towards its target and it isn't irrational. What I mean by this is that any secondary information that the company has received from an external source such as a survey, it is important to make sure that the secondary information is relevant to the company's needs and targets.

In basic terms, data from external sources are similar to the game Chinese whispers, if you're in a circle and you tell a person one thing which is then passed around the circle, it will come back to you as something completely different.

This can happen to data and can affect data quality. When the local replication, maintenance and redistributing copies of data occurs it can increase errors. Similarly, generating data from different sources can create data errors. Data should only ever be generated once, but can be used by many systems after that.

With the data having an accuracy of source, deciding where to get the information from is another way to improve the quality of information. To ensure that this information is relevant, they have to make sure it is comes from a reliable source.

Data collected from a primary source is often the most reliable and valid data making it quality information, it just needs to be analysed and the required information located and used. To ensure that this information is relevant, they have also to make sure it is comes from a reliable source.

Primary sources are the business functional areas and any research carried out by the teams themselves. To ensure that this information is relevant, they will again have to make sure it is comes from a reliable source.

Quality information can also come from secondary sources, but the company has to be careful when using this type of information. To ensure that this information is relevant, they will yet again have to make sure it is comes from a reliable source. However, not only that but, with secondary sources, the organisation will need to check whether the review is based on the organisation's needs.

Overall, it is important to make sure that the external information is reliable and beneficial towards the company. The reason being that unreliable data can be very time consuming and sometimes costly towards a company that has relied on it, especially if money was invested on unreliable data.

That information is transferred securely and timely around the system

In the company, if the flow of information was not very good. There will need to be a more efficient flow of information from the program or project managers to the rest of the team, and between the different sub-teams. Improving the flow of information within the company can be simple and there are several ways that it can be done.

Firstly, the way the company communicates to the each team will be different as they will be working in different areas of the college and on different parts. this means that method of passing on the information needs to be specific to that team. The type of information each team will require will be different as well, this can be a deciding factor in what type of communication to use when passing on information to a specific team.

When a form of communication has been established for each team, whoever is sending the information needs to make it specific to the recipients purpose. This is done by removing any unnecessary information that will not be beneficial to them. Removing unnecessary information will improve the flow of information because it will be more direct and useful for both the sender and receiver of the information.

Once the information needed has been found, it can be simplified and streamlined so that it is easier for the recipient to use. Simplified information will be easier to interpret and implement into the project. By streamlining the information, the information is more efficient to use.

An example of where information flow could be improved would be between the project managers and their teams. The project managers did not pass information down to their teams in the project so the teams were unaware of what they were doing throughout most of the project.

In the virtual security team, the project manager had a project plan for the group but did not give it to anyone else in the team so we had to rely on him to delegate tasks. Thus when he was away at meetings, the team did not complete any work. Improving the flow of information would solve this issue.

Overall, it is important that the information is transferred securely and timely around the system because the data is valuable towards a business due to its importance in improving the profitable gains and other beneficial aspects of a business. Having a good security and speed of the transfer is important so it can be safe from being intercepted and that it can be used as quickly as possible without it being interrupted.